From official statistics to sound: testing each link in a data-to-music pipeline for South Korean places

[Author names and affiliations pending] · Draft of 25 October 2026 · Project: Moodscape (sound-moodscape)

Status. Research questions RQ1–RQ5 are answered with project data and audio measured for this report. RQ4 includes an informal blind tap-tempo diagnostic by one listener for one Sejong clip; it is not a formal participant study and is not interpreted as an RQ6 result. RQ6–RQ8 (formal listener perception, in-situ validity, accessibility for intended users) have prepared protocols and **no data**. Entries marked † in the references still need verification.

Background. Moodscape turns official statistics about South Korean regions into music, so that a listener can sense a place's character without seeing a map. The idea rests on a chain of links, from regional data to a score, a musical specification, generated audio, and finally what people hear and what places are like. Each link can fail on its own. **Methods.** We tested the links one at a time, as research questions of the form “how does X lead to Y?”. We used nine official inputs for 17 regions, 75 landmark scores, a deterministic rule set for tempo, key and style, and 20 tracks generated with Suno in two rounds, measured with librosa. We additionally applied a blind tap-tempo diagnostic by one informal listener and deterministic post-processing to one Sejong clip. A Gemini-based style predictor is implemented and constrained to the data mapping; its comparison with the rule-based path is a hand simulation because no key-backed output is recorded. Formal listener ratings and field observations are specified but not yet collected. **Results.** The Activity Proxy ranks Seoul, Busan and Jeju highest, but the ranking is fragile to the stated weights (median $\rho = 0.87$ under modest perturbation; 41.8% of draws kept $\rho \geq 0.9$). Regions explain only 12.2% of landmark-score variance, so place-level sound mostly reflects authored offsets. Mode was matched in 5/10 and 7/10 clips (rounds 1 and 2), and none of the five minor-key requests in the two rounds returned minor, although some follow-up generations did. The requested tempo appeared among the strongest rhythmic peaks in 7/10 and 9/10 clips against about 1.5 expected by chance. However, Sejong's 95 BPM request was tapped at 48 BPM despite an automatic estimate of 95.3 BPM. A stricter prompt and a deterministic 95 BPM overlay also failed at 48 and 47 BPM; pitch-preserving retiming from 47 to 95 BPM plus replacement of the conflicting hi-hat produced a blind tap of 96 BPM (1.1% error). Changing genre words, not mode wording, flipped one clip from 0/5 to 3/3 minor but failed to transfer to a second clip. **Conclusion.** The first four links are implemented and measurable; two are weaker than the interface implies. The Sejong case shows that an automatic BPM match does not guarantee the pulse a listener perceives, and exact perceptual tempo may require deterministic post-processing rather than prompting alone. Formal listener and accessibility studies remain unrun.

Keywords: soundscape · sonification · ISO 12913 · generative music · South Korea · accessibility · validation

Introduction

Maps communicate difference between places through color and labels. Sound could carry some of the same information, and could do so for people who cannot use the visual channel. Moodscape is an interactive prototype that makes this claim concrete for South Korea: every region and landmark has a computed “mood”, the mood determines a musical specification, a generative-music service renders it as audio, and the interface exposes the same parameters as text and keyboard-navigable controls.

A prototype like this is a chain, and a chain is only as informative as its weakest link. A listener who hears a slow, minor-key track for a place is entitled to ask whether (i) the statistics really make that place calm, (ii) the specification really requests slowness and a minor key, (iii) the audio really is slow and minor, (iv) listeners hear calm, and (v) the place really sounds and feels calm. In use these are easy to blur into one claim. Here they are separated, and each is given its own question, method and evidence class.

Why separate soundscape perception from a computed score

Soundscape research defines a soundscape as the acoustic environment as perceived by people in context [2] and measures it with questionnaires and acoustic measurements, typically organized around two perceptual axes, Pleasantness and Eventfulness, arranged on a circumplex [3,4,5]. Oberman et al. [1] applied this framework in

protected natural areas, collecting 443 questionnaires at 28 evaluation points and showing that perceived sound-source dominance and psychoacoustic measures predicted perceived Pleasantness and Eventfulness better than sound pressure level alone. Two lessons carry over to this project. First, a single decibel-like or "activity" number is a weak description of what a place is like to hear. Second, the claim "this indicator corresponds to perceived Eventfulness" must be tested against in-situ perception, not assumed. Moodscape's Activity Proxy is not an ISO coordinate; it is an official-statistics composite that the field protocol described in Methods is designed to test.

The report follows the structure of [1]: research questions are stated up front, results are organized by question, methods follow the discussion, and descriptive tables, circumplex-style scatter plots and stacked bars are used where the data suit them.

Sonification for accessible exploration

Sonification of geographic data for users with visual impairment has precedent in coordinated speech and non-speech audio in map interfaces [6]. The project's design follows two principles: audio should carry data and not only decoration, and the interface should cooperate with the user's own screen reader (a correction made after an earlier version competed with it). Whether Moodscape's interface achieves this for blind or low-vision users has not been tested (RQ8).

Research questions

Each question follows one link of the chain in Fig. 1 and is phrased as a scenario.

- **RQ1.** How do official regional statistics lead to an Activity Proxy and a Visitor Pleasantness composite, and how stable is the resulting ordering?
- **RQ2.** How do regional values lead to landmark-level scores, and how does online attention relate to them?
- **RQ3.** How does a score lead to a musical specification (tempo, pitch, mode, style), and which rule decides each choice?
- **RQ4.** How does a specification lead to generated audio, i.e. does the audio have the requested tempo and mode, and what changes that?
- **RQ5.** How do the AI components (Gemini style prediction, language-accessibility estimate, and the generation request itself) lead to different or better specifications?
- **RQ6.** How does generated audio lead to listener-perceived Eventfulness? *(Prepared; no data.)*
- **RQ7.** How does the model relate to perception in the actual places? *(Prepared; no data.)*
- **RQ8.** How does the interface lead to independent, accessible exploration? *(Implemented; not user-tested.)*

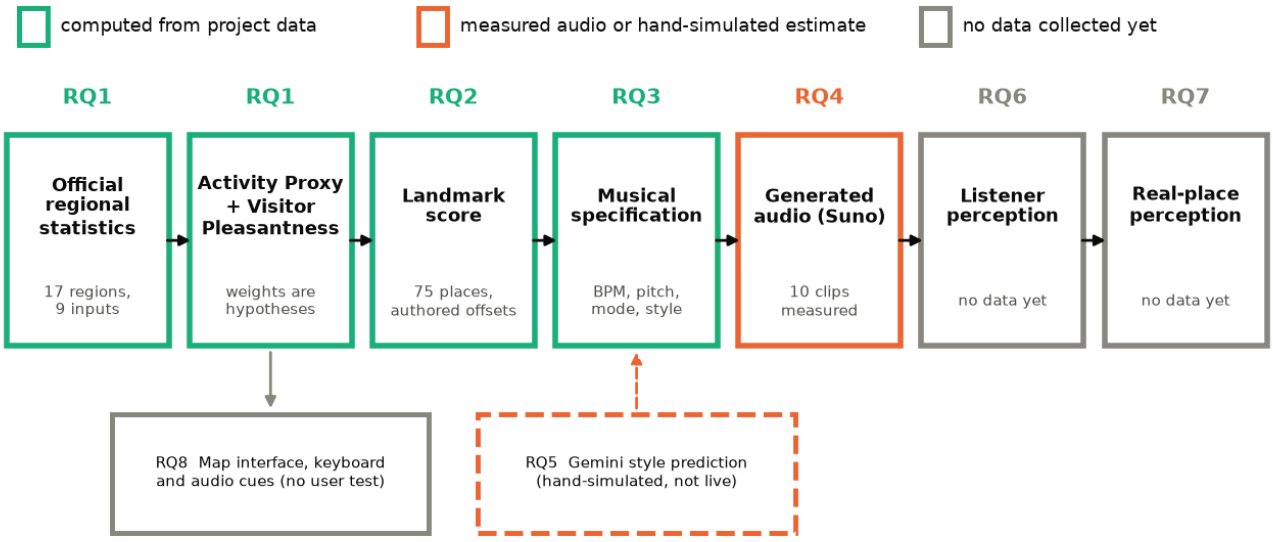


Fig. 1. The chain tested in this report. Each box is one link and carries the research question that tests it. Green outlines mark links computed from project data, orange marks measurement on generated audio or a hand-simulated (not live) estimate, gray marks links for which no data have been collected. Dashed boxes are branches: the Gemini style predictor (RQ5) feeds the musical specification; the interface and accessibility cues (RQ8) read from the score.

Terms used in this report

Requested	What the prompt asked for (a tempo in BPM, major or minor), taken from the data mapping.
Detected (“measured”)	A value estimated by software from the downloaded audio. It is not a human judgment and not a reading taken in a place.
Key-fit score	The correlation between a clip's pitch-class profile and the best-fitting major or minor key profile. Higher means a clearer fit. It is not the probability that the key is correct.
Round 1, round 2	Two separate batches of ten newly generated clips (new requests to the provider, new audio): round 1 with the original prompt format, round 2 with the revised format. “Follow-up” generations are single extra generations for particular landmarks.
Matched / honored	The detected value agrees with the requested one (same mode; tempo within the stated tolerance).
Half/double-tempo equivalent	A detected tempo divided or multiplied by two, whichever lands nearest the request. It is another way of comparing the same detected number: the audio, the tempo formula and the detection are unchanged. Because it uses the request, it can only shrink the gap.
Requested tempo “present”	The requested tempo lies within 5% of one of the three strongest rhythmic periodicities found in the audio.
Illustrative score	A landmark score set by the authors as an offset from its regional value; not measured.
Fixed inputs	Tempo, key, root note and vocal treatment supplied by the data mapping, which the language model may not change.
Hand-simulated	A prediction produced by a researcher following a function's instructions by hand on real photos and text; not output from the live model.
Custom request shape	The provider's request format with separate style and title fields, as opposed to one free-text prompt.

Results

Results are reported for each research question in chain order. Evidence classes: **computed** (from project data and code), **measured** (on generated audio), **hand-simulated** (a researcher followed a function's instructions by hand; not output from the live model), **pending** (no data). Unless stated otherwise, intervals are 95%, tests are two-sided, and permutation tests use 20,000 permutations (seed 20260920).

RQ1 — How do official regional statistics lead to an activity score?

> *Scenario*. A visitor opens the map. Each of 17 regions has a color and a tone. Where does that value come from, and would a reasonable change of assumptions give the same picture?

Inputs and outputs. The Activity Proxy ranged from 32.9 to 61.3 (mean 46.4, SD 8.0; Table 1, Fig. 2a). Seoul, Busan and Jeju-do scored highest and Gwangju lowest. The Visitor Pleasantness composite occupied a much narrower band, 75.0–81.7 (SD 2.1), highest in Jeollanam-do and Gwangju and lowest in Daejeon and Sejong (Fig. 2b).

Table 1. Descriptive statistics of the 17 regional inputs and the two composites. Values are computed from the nine inputs; the model is described in Methods.

Measure	Min.	Max.	Mean	Median	SD
Population density (per km²)	90	15432	2101	855	3671
Domestic tourism trips per resident	0.94	20.48	6.90	3.85	5.72
Noise/vibration complaints per 100,000 residents	77	679	242	212	148
Regulated noise-emitting facilities per 100,000 residents	3	214	72	38	70
Crowding pressure (100 – crowding satisfaction)	17.90	29.30	22.88	23.00	3.01
Overall satisfaction (0–100)	77.80	83.30	80.33	80.00	1.90
Recommendation intention (0–100)	71.00	79.30	74.86	74.80	2.27
Revisit intention (0–100)	72.60	81.60	76.86	76.90	2.39
Crowding satisfaction (0–100)	70.70	82.10	77.12	77.00	3.01
Activity Proxy (0–100)	32.91	61.27	46.42	44.67	7.98
Visitor Pleasantness composite (0–100)	75.05	81.74	77.79	77.72	2.11

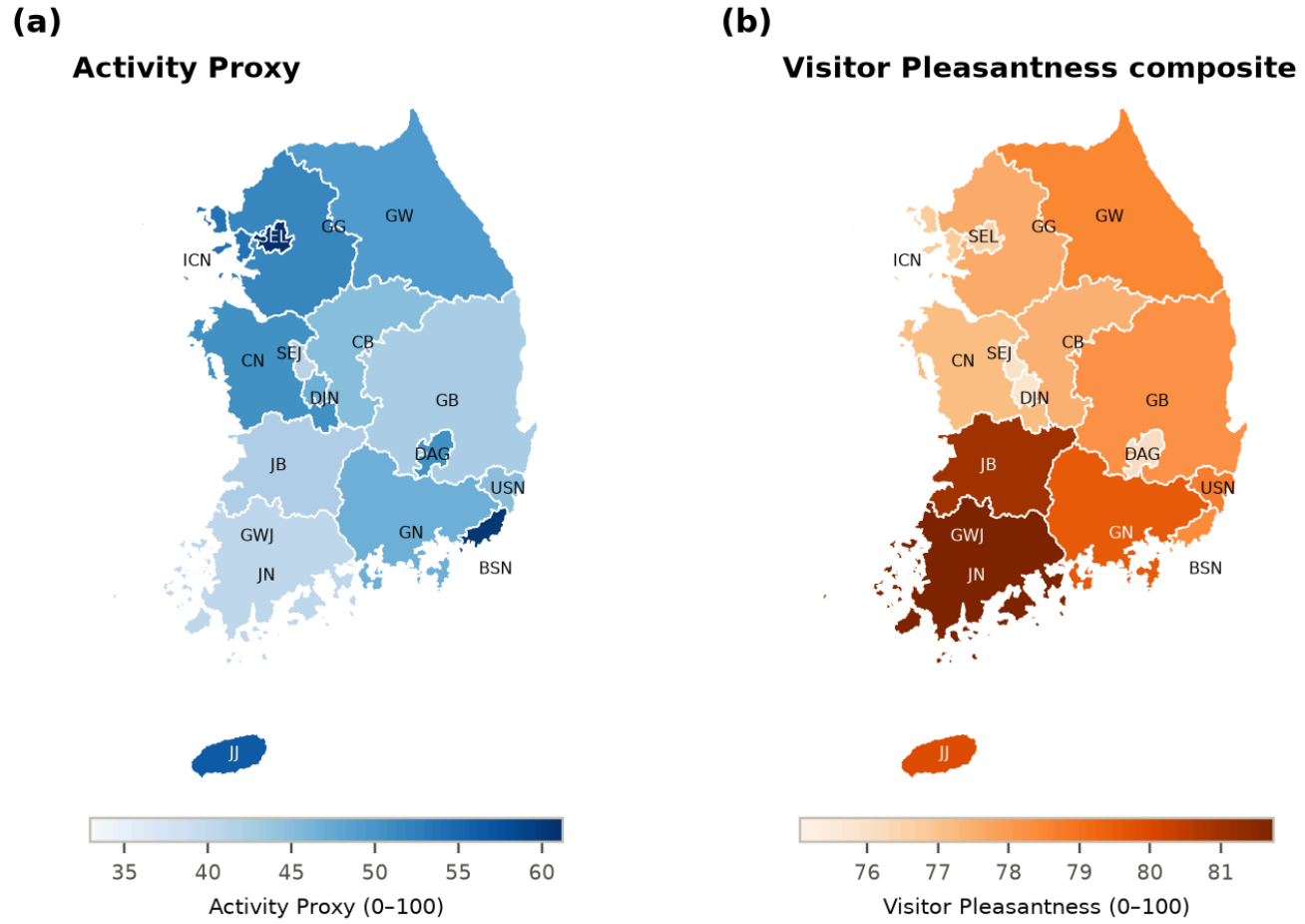


Fig. 2. South Korea by region. **(a)** Activity Proxy (0–100; darker = more active). **(b)** Visitor Pleasantness composite (0–100; darker = higher). Region abbreviations: SEL Seoul, BSN Busan, DAG Daegu, ICN Incheon, GWJ Gwangju, DJN Daejeon, USN Ulsan, SEJ Sejong, GG Gyeonggi, GW Gangwon, CB/CN Chungcheongbuk/nam, JB/JN Jeollabuk/nam, GB/GN Gyeongsangbuk/nam, JJ Jeju. Boundaries: `geojson.js`.

Effective influence differs from stated weight. The weights (0.35, 0.30, 0.15, 0.10, 0.10) are design hypotheses. The rank association of each component with the final score did not follow them (Table 2): complaints ($\rho = 0.68$) and crowding pressure (0.65) tracked the score closely, whereas tourism intensity, despite a weight of 0.30, did not (-0.07). Removing density, the heaviest component, reduced agreement with the full ordering to $\rho = 0.23$; removing any other single component left $\rho \geq 0.60$. Equal weights reproduced the ordering with $\rho = 0.94$.

Table 2. How each component relates to the Activity Proxy. ρ (component) is the Spearman correlation of the scaled component with the final score across the 17 regions. ρ (drop one) is the agreement of the ranking after removing the component and renormalising the other weights.

Component	Transform	Weight	ρ (component)	ρ (drop one)
Population density (per km²)	log1p, min–max	0.35	0.45	0.23
Tourism trips per resident	log1p, min–max	0.30	-0.07	0.60
Noise complaints per 100,000	log1p, min–max	0.15	0.68	0.85
Noise-emitting facilities per 100,000	log1p, min–max	0.10	-0.20	0.91
Crowding pressure (100 – satisfaction)	log1p, min–max	0.10	0.65	0.94

Robustness to weights. We drew 10,000 weight vectors from a Dirichlet distribution centered on the stated weights (concentration 20; weight SDs 0.07–0.10) and, as an extreme, 10,000 from a flat Dirichlet. Under modest perturbation the median agreement with the stated ordering was $\rho = 0.87$, but the 5th percentile was 0.34 and the minimum 0.00; only 41.8% of draws kept $\rho \geq 0.9$ (flat draws: median 0.72, 14.5%; Fig. 3b). The median 95% interval of a region's rank spanned 9 of 17 ranks (Fig. 3a). Only Seoul (74%), Busan (90%), Jeju-do (56%) stayed in the top three in more than half of the draws.

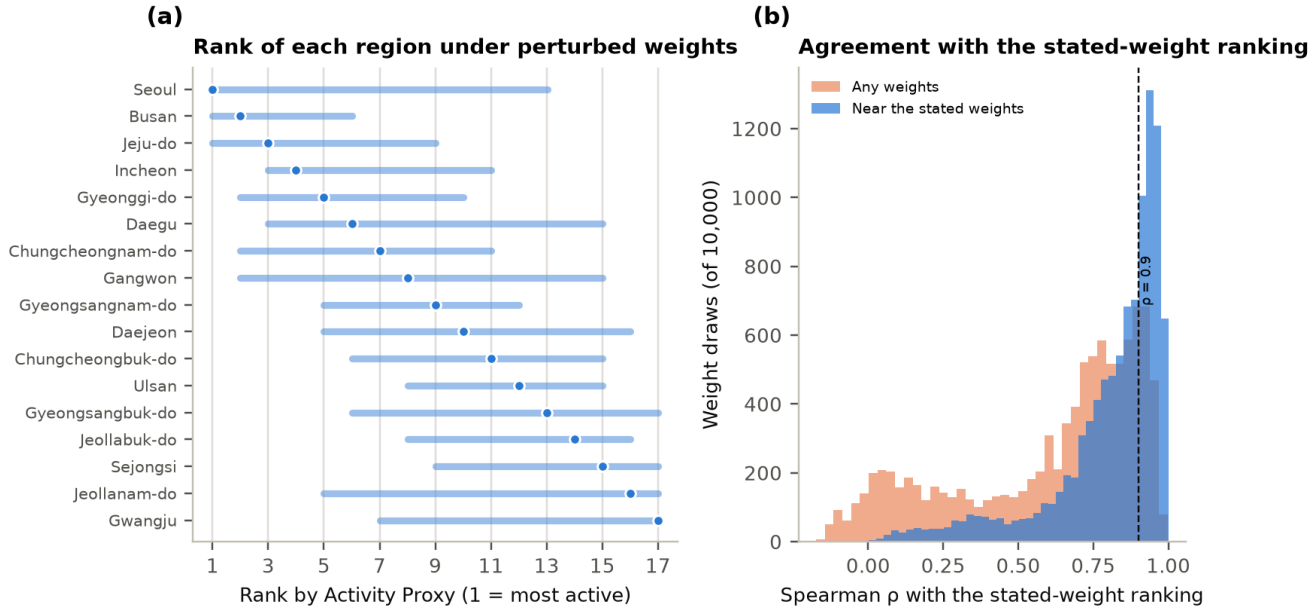


Fig. 3. Sensitivity of the Activity Proxy to its weights. **(a)** Rank of each region under the stated weights (dot) and its 95% range across 10,000 weight vectors drawn near the stated weights (bar). **(b)** Distribution of Spearman ρ between each perturbed ranking and the stated-weight ranking, for weights near the stated values (blue) and for arbitrary weights (orange). Dashed line: ρ = 0.9.

The only external reference. Two regions have directly published soundscape studies. The model places Seoul first (score 61.3) and Daejeon 10th (44.6), the same order as their published Eventfulness values (78 and 45). Two points can show a contradiction but cannot validate a model; this check is a sanity test only.

RQ2 — How do regional values lead to landmark scores?

> *Scenario*. The visitor opens Hongdae and Gyeongbokgung Palace, both in Seoul. Do they sound different because the data say so, or because someone decided so?

Region explains little of the landmark score. Each of the 75 landmarks has a score equal to its region's Activity Proxy plus an authored offset, clipped to [0, 1]. The offsets had SD 0.21 and ranged from -0.67 to +0.40 (score units 0–1), compared with a regional SD of 0.08. In a one-way analysis of the landmark scores by region (3–6 landmarks per region), region explained 12.2% of variance (η^2 = 0.122; ICC(1) = -0.128, which is indistinguishable from zero). 3 landmarks were clipped at the floor of 0, so their authored calmness is partly hidden. The place-level score, and therefore the place-level sound, reflects authored offsets far more than official statistics. The interface labels these scores *illustrative*; no landmark is marked as measured.

Online attention. A separate, reproducible readership index, the log-scaled monthly English-Wikipedia pageviews over a common 12-month window (1 Sep 2025–31 Aug 2026), was available for 58 of 75 landmarks (16 had no usable article and 1 was incomplete). It was weakly and positively associated with the landmark score (Spearman ρ = 0.25, permutation p = 0.062, n = 58; rank plot in Fig. 7c), which does not distinguish it from zero at the 5% level. Pageviews measure online attention to an article, which is sometimes broader than the landmark (for example a company or city article), and are not footfall, noise or perceived Eventfulness; the index never alters the music.

RQ3 — How does a score lead to a musical specification?

> *Scenario*. The score for a place is known. What exactly is asked of the music, and which rule decides each request?

Tempo and pitch. Tempo is *round(42 + 110 × score)* BPM, giving 42–138 BPM across the 75 landmarks (the formula allows 42–152 BPM; the landmarks reach 42–138). Pitch is a chromatic step of *round(12 × score)* semitones above A4 (440–880 Hz); the music-specification panel reports it, the navigation tone uses a related penta-

tonic mapping stretched across the observed score range, and its letter name is passed to the generator as a soft root-note hint (“around F”).

Mode is decided by three rules, and the one tied to Visitor Pleasantness decides less than half of the places. 40 of 75 landmarks (53.3%) receive their mode from a category override (6 minor for solemn places, 34 major for traditional, nature and administrative places). The remaining 35 (46.7%) receive it from the Visitor Pleasantness default: major if the region's composite is at or above the median of the 17 regions (77.72), minor otherwise (Table 3, Fig. 4). Overall 21 landmarks (28.0%) request a minor key.

Table 3. Which rule decides the requested mode, current code, 75 landmarks. Rules are applied in the order listed, so a place that matches an earlier rule is not counted under a later one.

Deciding rule	Major	Minor	Places
Solemn override (memorial, cemetery, observatory, cave, …)	0	6	6
Traditional/nature override (hanok, temple, palace, park, beach, government, …)	34	0	34
Regional Visitor Pleasantness default (vs. median of 17 regions)	20	15	35
Total	54	21	75

Counts differ by one place from an earlier internal count (35/34) written before later edits to the category lists; this table uses the model as of 11 Oct 2026.

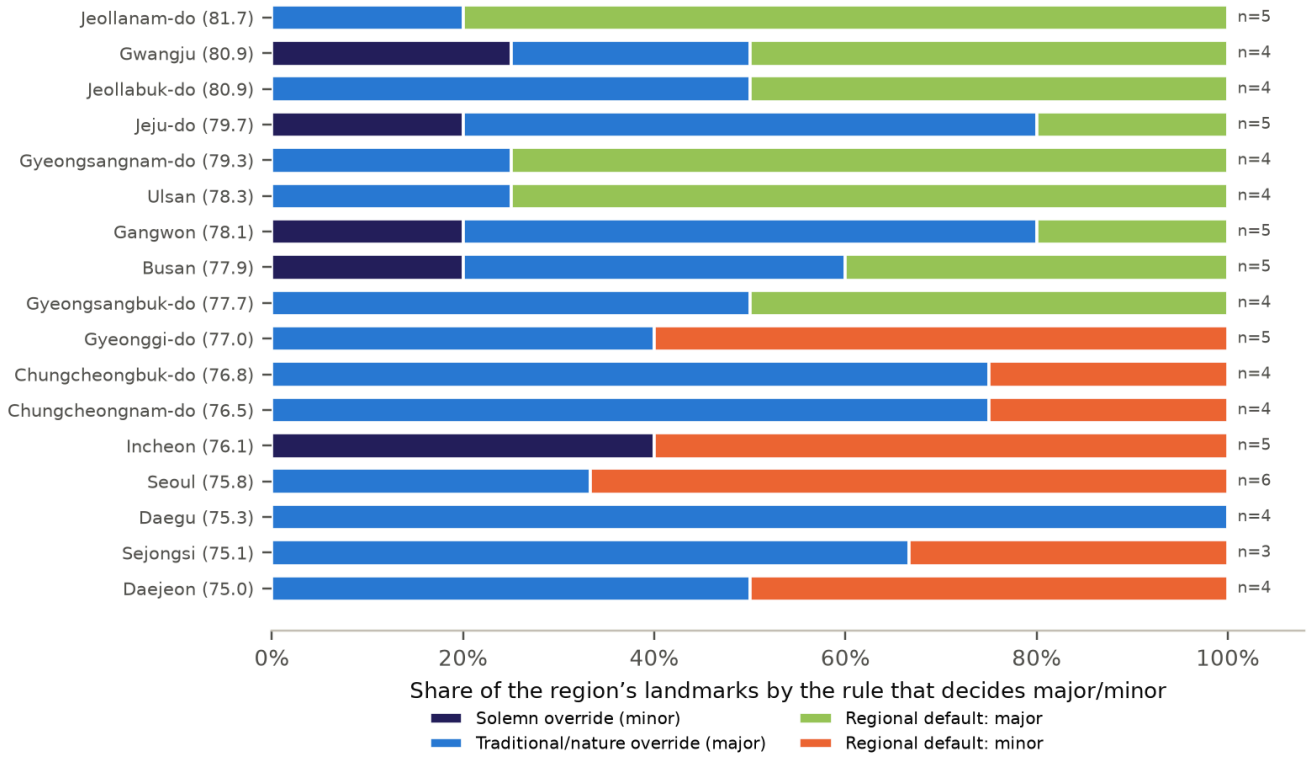


Fig. 4. Share of each region's landmarks by the rule that decides major or minor. Regions are ordered by their Visitor Pleasantness composite (value in brackets, highest at top). Solemn and traditional/nature overrides ignore Pleasantness. Only the green and orange segments depend on it.

The Pleasantness threshold sits inside the noise. The 17 composites span 75.0–81.7, a band of 6.7 points. The cut-off is the sample median, so by construction about half of regions (8 of 17: Seoul, Daegu, Incheon, Daejeon, Sejongsi, Gyeonggi-do, Chungcheongbuk-do, Chungcheongnam-do) receive minor for their default places, regardless of how pleasant all regions are in absolute terms. 6 regions lie within one point of the median, and Gyeongsangbuk-do sits exactly on it and is classified major only because the comparison is ≥. Because the value is regional, every default-branch place in a region shares one decision: Hongdae, Gangnam, N Seoul Tower and Dongdaemun Design Plaza are all minor for the same reason. There are 17 distinct Pleasantness decisions, not 75 (Fig. 5).

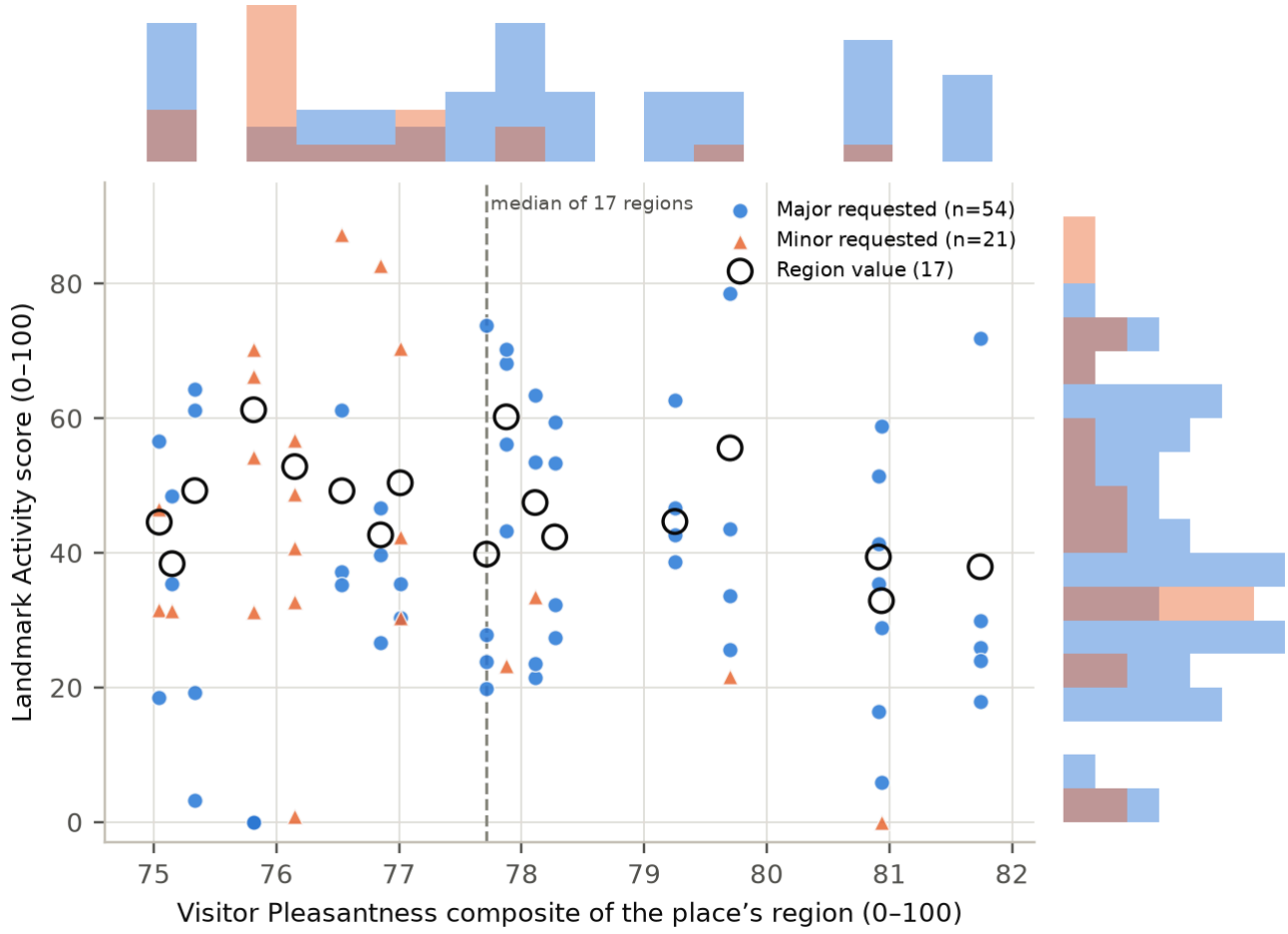


Fig. 5. The modeled space, for visual comparison with the ISO circumplex in [1] (not ISO coordinates; both axes are modeled proxies). Each point is a landmark: horizontal position is its region's Visitor Pleasantness composite, vertical position is its own Activity score. Blue circles requested major, orange triangles requested minor; open circles are the 17 region values. Marginal histograms are shown on the top and right. The dashed line is the median that decides default mode. Landmarks sit in vertical columns because Pleasantness is a regional value.

Style words and mode. Genre and mood words are chosen by place category and score band, independently of the mode; the mode is then stated in an early sentence. For a minor key the sentence reads, in effect, "in a minor key (around *root*) — melancholic, somber, shadowed tonality, no major-key brightness", and for a major key "in a major key (around *root*) — bright, open, uplifted tonality". An earlier version let positive-valence words such as "warm" and "mellow" accompany a minor request. The current rule removes a fixed list of such words when the mode is minor. 12 of the 21 minor-key styles still contain at least one of "relaxing", "peaceful" or "serene" (an exploratory keyword count, not a validated measure of valence). Whether mode is the right carrier for visitor satisfaction was not tested; it is a design assumption borrowed from the common association of minor mode with lower valence.

RQ4 — How does a specification lead to generated audio?

> *Scenario*. The prompt says "minor key, around 115 BPM". A generative model returns a track. Is it minor, is it near 115 BPM, and what would change that?

Round 1 (original prompts, non-custom request). Ten landmarks spanning the observed score range (0–87) were generated once each. The requested mode was detected in 5/10 tracks (95% Clopper–Pearson interval 19–81%): 5/7 of the major requests and 0/3 of the minor requests (Fig. 6b, Table 4). Mean key-fit score was 0.89 for matches and 0.74 for mismatches; the Gangnam miss (A♯ major) was detected with key-fit score 0.92, so the detector was not undecided. Tempo agreed within ±10% in 2/10 tracks, and the rank correlation between re-

quested and measured tempo was ρ = 0.07 (p = 0.864). Nine of ten measured tempos fell between 114 and 161 BPM although the requests ranged from 42 to 138 BPM.

Round 2 (revised prompts, custom request). After the prompt builder changed (mode moved to an early sentence with a root note, positive-valence words removed from minor styles, an administrative-building category added, request switched to the provider's custom mode), the same ten landmarks were generated again with new requests to the provider (new audio). The requested mode was detected in 7/10 (95% interval 35–93%): 7/8 major requests and 0/2 minor requests. Tempo agreed within ±10% in 4/10 tracks (ρ = 0.08, p = 0.829). One of the three round-1 minor requests (Sejong) became a major request in round 2 after a factual correction, so the two rounds do not test the same requests; the improvement from 5/10 to 7/10 is partly an artifact of that change and its interval overlaps round 1's.

Table 4. Requested and measured tempo and mode for the ten landmarks in both rounds. Measured tempo is the beat-tracking estimate (see Methods); ratio = measured / requested. ✓ = requested mode detected; ✗ = other mode. Round 2 asked for a major key at Sejong (clip 07); round 1 asked for minor.

Clip	Landmark	Score	Req. BPM	Round 1 BPM	Round 2 BPM	Mode asked	Round 1 key	Round 2 key
01	Gyeongbokgung Palace	0	42	122 (2.91×)	131 (3.11×)	major	D# major ✓	A major ✓
02	Hanbat Arboretum	19	62	125 (2.02×)	125 (2.02×)	major	D# major ✓	B major ✓
03	Naganeupseong Folk Village	24	68	131 (1.92×)	131 (1.92×)	major	G major ✓	C major ✓
04	Korean Folk Village	30	75	152 (2.03×)	75 (1.00×)	major	D# major ✓	C# major ✓
05	Oeam Folk Village	35	81	161 (1.98×)	161 (1.98×)	major	B minor ✗	C# major ✓
06	Gunsan Modern History Street	41	88	44 (0.50×)	60 (0.68×)	major	C major ✓	B minor ✗
07	Government Complex Sejong	48	95	128 (1.35×)	95 (1.00×)	minor	D# major ✗	D# major ✓
08	Kia AutoLand Gwangju	59	107	144 (1.35×)	144 (1.35×)	major	G minor ✗	B major ✓
09	Gangnam	66	115	115 (1.00×)	117 (1.02×)	minor	A# major ✗	G# major ✗
10	Hyundai Motor Asan Plant	87	138	137 (0.99×)	137 (0.99×)	minor	D major ✗	D major ✗

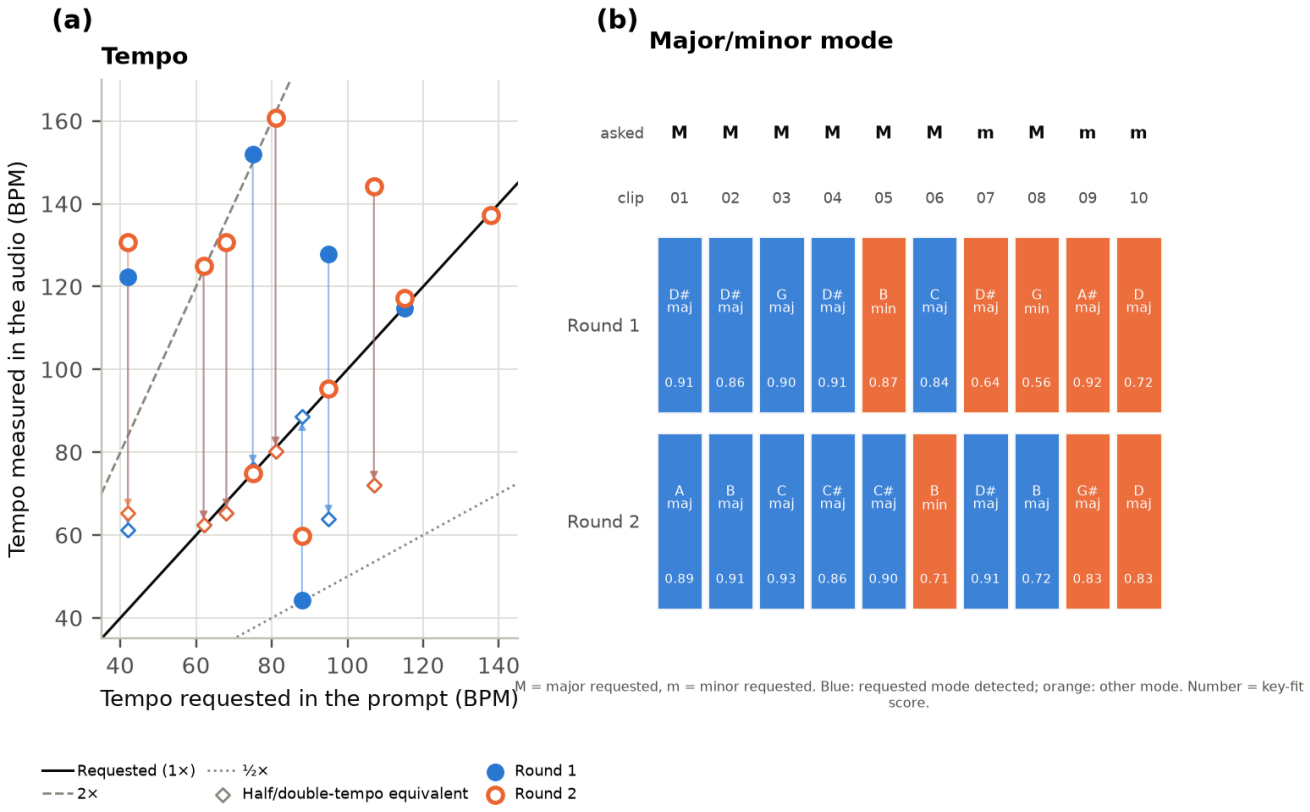


Fig. 6. Generated audio against the specification. **(a)** Requested vs measured tempo for round 1 (filled) and round 2 (open). The solid line is exact agreement; dashed and dotted lines mark double and half tempo. Each arrow ends at a small diamond, the half- or double-tempo equivalent of that detected value, whichever lies nearest the request. This is a different way of comparing the same detected number; the tempo formula, the audio and the detection are unchanged. Because the choice uses the request, it can only shrink the gap, so it shows the requested pulse is plausible at another metrical level, not that the audio is at the requested tempo. Clips whose diamond is still far from the solid line (for example 01, 07 and 08 in round 1) are not explained by a half or double reading. **(b)** Detected key for each clip and round. Blue: requested mode detected; orange: other mode. The number is the key-fit score.

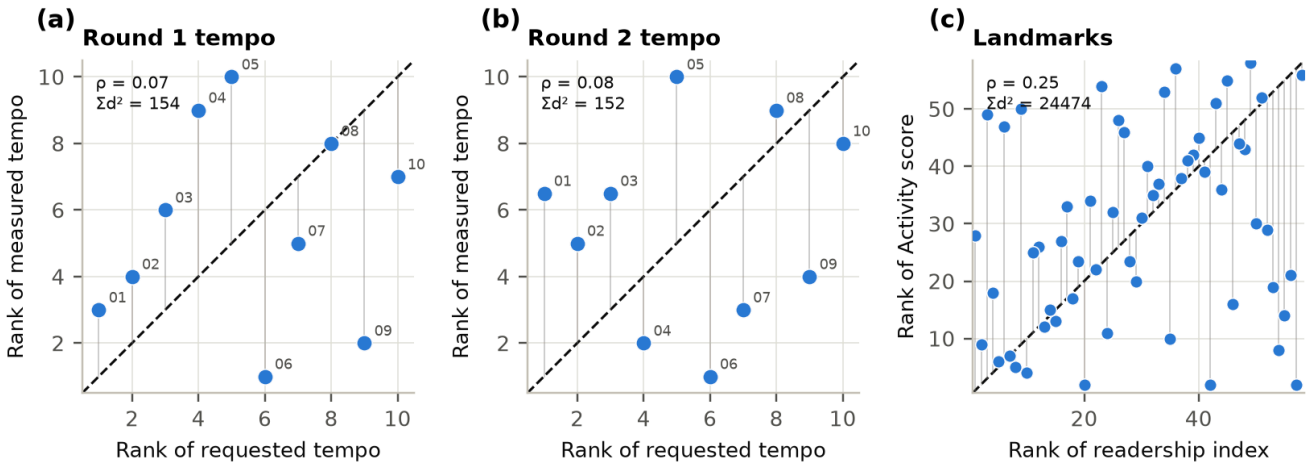


Fig. 7. Spearman rank agreement in three real analyses. Each point is one item. Its horizontal position is its rank on one variable and its vertical position its rank on the other (1 = smallest). The dashed black diagonal is perfect agreement (ρ = +1). The orange line is the average trend of the points: with ranks, its slope equals ρ, so a flat line means no relationship and the diagonal means a perfect one. The thin vertical segment from a point to the diagonal is that item's rank difference d; Σd^2 is the sum of the squared differences, and without ties $\rho = 1 - 6\Sigma d^2 / (n(n^2 - 1))$. Tied values get average ranks, so some ranks are fractional. **(a)** Requested versus measured tempo for the ten clips, round 1. **(b)** The same for round 2. **(c)** Wikipedia readership index versus landmark Activity score for the 58 landmarks with a complete record. Points near the diagonal mean the two variables order the items alike; points scattered across both triangles mean they do not.

Is the tempo wrong or mis-measured? Rank agreement between requested and measured tempo is shown in Fig. 7a,b. A single tempo estimate can pick a different metrical level from the one a listener would tap. We test-

ed this with a prior-free view of the same audio: the three strongest peaks of each clip's tempogram. The requested tempo (±5%) was among them in 7/10 round-1 clips and 9/10 round-2 clips, against 1.5 and 1.6 expected if the requests were randomly shuffled among the clips (95th percentile 3 and 4; $p < 0.0001$ for both; Table 5, Fig. 8). The single best estimate, in contrast, was within ±5% in only 3 and 4 clips. Removing librosa's default tempo prior (centered at 120 BPM) did not rescue the estimate (round 1 $\rho = 0.09$ with the default prior, 0.22 with a flat prior). In round 1 the three slowest requests (42, 62 and 68 BPM) never appeared among the top peaks (0/3), while every request of 75 BPM or more did (7/7); in round 2 two of those three slow requests appeared (2/3), and only the 42 BPM request did not. So tempo is usually present but not usually the strongest rhythmic level, and the slowest request is the least reliably honored. This analysis uses the request when judging presence, so it shows that the requested pulse is *detectable*, not that a listener would perceive it as the main beat. 5 of ten single estimates were numerically identical across the two independent rounds (for example 125.0, 130.8, 137.2, 144.2 BPM), which is consistent with a discrete tempo grid in the estimator.

Table 5. Tempo diagnostic. "Single estimate" is the global tempo estimate with its default prior (log-normal, centered at 120 BPM); "top-3 peaks" is the number of clips with the requested tempo (±5%) among the three strongest autocorrelation peaks. Expected value (95th percentile) is from 20,000 shuffles of the requests among clips.

Round	Single estimate within ±5%	Top-3 peaks contain request	Expected by chance (95th pct)	Permutation p	Requests ≤ 68 BPM	Requests ≥ 75 BPM
Round 1	3/10	7/10	1.5 (3)	$p < 0.0001$	0/3	7/7
Round 2	4/10	9/10	1.6 (4)	$p < 0.0001$	2/3	7/7

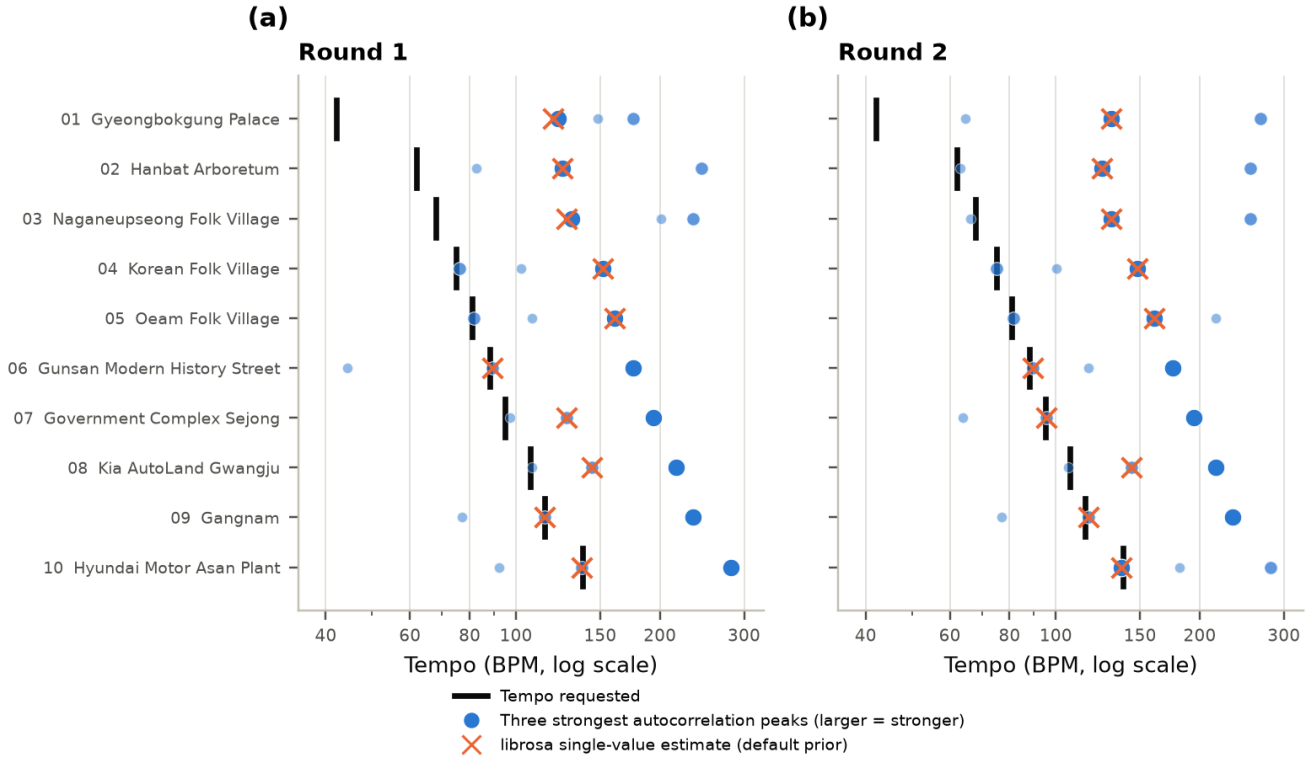


Fig. 8. Tempo diagnostic for each clip. The black bar is the requested tempo. Blue dots are the three strongest autocorrelation peaks of the onset envelope (larger = stronger). The orange cross is the single value returned by librosa with its default prior. A black bar that coincides with a blue dot that the cross misses means the requested pulse is present but not the dominant level. **(a)** Round 1. **(b)** Round 2.

The pulse a person heard: the 95 BPM Sejong case. On 25 October 2026, one informal listener tapped the main pulse of clip 07 without seeing its name or target (Table 6). The round-2 source requested 95 BPM and the default automatic estimate was 95.3 BPM, but the listener heard 48 BPM as the main pulse. A regeneration that explicitly required a 95 BPM quarter-note pulse and prohibited half-time was also tapped at 48 BPM. Overlaying

an exact 95 BPM kick and rim pulse on the source was tapped at 47 BPM, indicating that the added pulse was absorbed as a subdivision of the existing slow beat. Finally, the source was pitch-preservingly retimed by 2.021× from the human measurement of 47 BPM to the target of 95 BPM, shortening it from 164.413 to 81.341 seconds. Existing hi-hat energy above a 5.2 kHz shelf was reduced to 12%, and a new closed hi-hat plus kick and rim were placed every 0.6316 seconds. A new anonymous 25-second clip was tapped at 96 BPM, 1 BPM (1.1%) from the target and within the ±5% criterion used before the test.

Table 6. Perceptual-tempo diagnosis and correction stages for Sejong clip 07. Automatic values are librosa default estimates; human values are blind taps by the same one informal listener. The final row is post-processed audio, not a raw generation.

Stage	Target BPM	Automatic BPM	Human tap BPM	Decision
Round-2 Suno source	95	95.3	48	Fail
Regenerated with strict 95 BPM prompt	95	95.3	48	Fail
Source + deterministic 95 BPM pulse	95	95.3 (47.7 with flat prior)	47	Fail
47→95 pitch-preserving retime + hi-hat replacement	95	95.3	96	Pass (1.1% error)

This is a sequential engineering diagnostic at one place with one listener (n = 1), not an effect estimate or a generalizable success rate, and it is not part of the RQ6 listener experiment.

The case supports a narrower conclusion than “the automatic analysis was wrong.” The same audio can contain hierarchical periodicities near 95 and 48 BPM; an estimator with its default prior may choose 95 while a listener selects 48 as the main pulse. Therefore, a detected request or an automatic match does not establish *perceived tempo fidelity*. The final 96 BPM result demonstrates deterministic correction on one clip; it is not evidence that Suno's prompt fidelity improved.

What changes the mode? Genre words, in one place. The two clips that repeatedly failed to come back minor, Gangnam and the Asan plant, were studied with additional generations (Table 7, Fig. 9). Across 10 Gangnam generations, none of the 5 that kept the original genre words (“lo-fi hip-hop, city-pop”) came back minor, regardless of rewording, request shape or an explicit root note; 1 of 2 came back minor when no genre words were given; and 3 of 3 came back minor when the genre was swapped to “dark synth-pop, moody downtempo”. The same swap did not transfer: for the Asan plant, 0 of 3 swapped-genre generations were minor, and 1 of 3 with the original wording (the one success and a later failure used the same text). Where the root was requested, it was matched in one of three successful Gangnam minor tracks (detected F, G, D♯; requested F). The Sejong clip shows a different cause: a keyword collision gave a government building a semiconductor-plant style, and a factual correction (a planned city with many young families) changed the requested mode; both were defects in the project's own code or data and were fixed, with the generated audio matching the corrected request in the follow-up generations (D♯ minor with key-fit score 0.83 under the minor request; D♯ major with key-fit score 0.94 under the major request).

Table 7. Every generation for the two clips that failed repeatedly, grouped by what the prompt said about genre. Detected key (key-fit score). “Minor” counts the requested mode obtained. Tracks returned by the same API call are listed together and are not independent.

Prompt / request	Genre words	Detected key (key-fit score)	Minor / n
Gangnam			
Round 1 wording, non-custom	original genre words	A# major (0.92)	0/1
Reworded, non-custom	original genre words	D# major (0.94)	0/1
Explicit "A minor", custom	original genre words	C major (0.89), C major (0.86)	0/2
Round 2 prompt, custom	original genre words	G# major (0.83)	0/1
Mode + tempo only (no genre words)	no genre words	D major (0.70), E minor (0.69)	1/2
Genre swapped to dark synth-pop	swapped genre words	F minor (0.64), G minor (0.68), D# minor (0.81)	3/3
Asan plant			
Round 1 wording, non-custom	original genre words	D major (0.72)	0/1
Reworded, custom	original genre words	G minor (0.67)	1/1
Round 2 prompt, custom	original genre words	D major (0.83)	0/1
Genre swapped to dark techno	swapped genre words	D major (0.81), D major (0.59), G major (0.59)	0/3

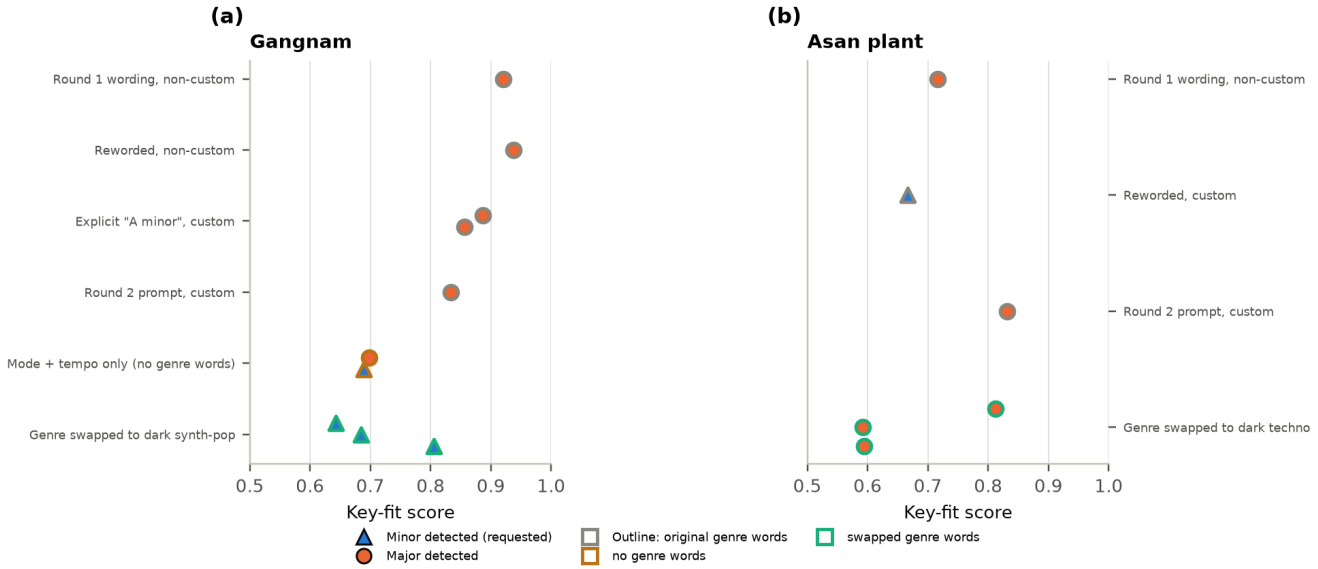


Fig. 9. Detected key and key-fit score for every generation of the two minor-key clips. Triangles: minor detected (requested). Circles: major detected. Outline color: gray = original genre words, brown = no genre words, green = swapped genre words. **(a)** Gangnam. **(b)** Asan plant.

These follow-up experiments were small, run post hoc, and partly by hand; they generate hypotheses, not estimates. Suno's output is stochastic: the same Asan prompt produced a minor track in one generation and a major track in another.

Why the prompt is built this way, and what raises the success rate. A generation request is a short text with four jobs: place the music in the data (tempo, mode, root note), say what it should feel like (a scene sentence, genre and mood words), say what it should sound like (instrumentation) and say what to leave out (vocals). The provider treats the text as a soft instruction and the output is random, so the useful question is which parts of the text raise the probability that a requested parameter appears. Table 8 gives the reasoning for each part and the evidence we have; Fig. 10 pools every minor-key request for which we have audio.

Table 8. Anatomy of the generation request: why each part is there, and what the data say about it. “Suggestive” means a difference in the expected direction that is not distinguishable from chance at this sample size.

Part of the request	Reasoning: why it should help	Evidence in this study	Status
Separate style and title fields (the provider's custom mode)	The provider documents the style field as an explicit control, so the text is read as music settings, not a scene to reinterpret.	Minor obtained in 8/15 custom requests vs 1/6 non-custom (p = 0.178).	Suggestive
Mode stated early, with affect words and a root hint	An early position gives the mode priority over later style words, and affect words (“melancholic, somber”) repeat the cue in the vocabulary the model is most likely to follow, which a bare theory term does not.	6/12 with the descriptive sentence vs 3/9 with a bare term (p = 0.660).	Suggestive
Remove positive-valence words when the mode is minor	Words such as “warm” and “mellow” carry major-mode associations and compete with the mode request.	Round 2 applied this and Gangnam still failed, because the genre nouns remained; removal alone is not sufficient.	Necessary, not sufficient
Genre and style nouns	Genre names carry strong priors from the model's training data (for example lo-fi and city-pop sound major), so they can outvote a single mode phrase.	Gangnam with swapped genre words 3/3 vs 1/7 otherwise (p = 0.033, uncorrected); Asan 0/3.	Hypothesis; does not generalize
Tempo as “around *T* BPM”	A number is concrete and the provider can follow it approximately; “around” avoids promising precision.	Requested tempo among the three strongest rhythmic peaks in 7/10 and 9/10 clips vs about 1.5 by chance; weakest for the slowest request.	Supported, imperfect
Root note as a soft hint	A concrete anchor may nudge the tonal center; stated as “around” because exact control is not expected.	Matched in 1 of 3 successful Gangnam minor tracks.	Not reliable
Fixed parameters in the AI-assisted prompt	The data mapping fixes tempo, mode and root; the language model supplies only genre, instrumentation, mood and arrangement, so it cannot contradict the mapping or add an unrequested tempo word.	Design property; no live output recorded.	By construction
Category rules and real-world facts	The style and mode must describe the place correctly before any wording can matter.	Fixing a keyword collision and one missing fact changed Sejong's outcome cleanly.	Supported

Pooled evidence. Across all 21 minor-key requests with audio, 9 (42.9%) returned minor. Splitting by what the prompt contained (Fig. 10), no single feature separated success from failure at this sample size, and the features are confounded: every genre swap used the custom shape and the descriptive wording. The one contrast that stands out is Gangnam, and it is exploratory (one of several comparisons, uncorrected). A simple account fits the pattern: the mode words and the genre words are competing cues, and the genre cue wins when it carries a strong prior. That account predicts the Gangnam result and not the Asan result, so it is a hypothesis to test with more landmarks.

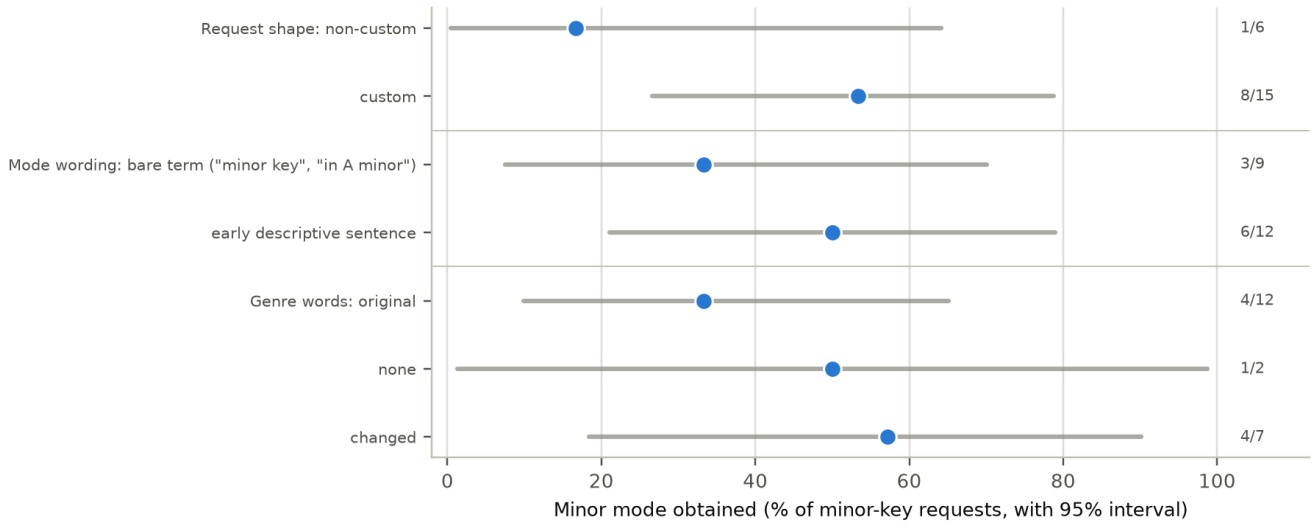


Fig. 10. Success rate of minor-key requests by what the prompt contained, pooled over every generation with audio (21 tracks). Dots are the share of requests that returned minor, bars are 95% Clopper–Pearson intervals, and the fraction on the right is successes/requests. The groups within a block overlap in time and design, so the rows are not independent experiments. “Genre words: changed” includes the Gangnam and Asan genre swaps and the Sejong category fix.

How the success rate can be raised. Three levers follow from the reasoning and the data. First, *remove cues that contradict the request*: this was the clearest gain, because it repairs the project's own category logic and data and costs no generation credits. Second, *phrase the request the way the provider documents it*: separate style field, mode stated early in affect words, tempo and root as soft hints, and genre words tested per landmark and not assumed to transfer. Third, *generate and check*: because output is random, detecting the key automatically and regenerating on a miss turns a per-track success rate p into a success probability $1 - (1 - p)^m$ over m tracks. With $p = 0.33$ (the rate with the original genre words), one call returning two tracks succeeds with probability at most 0.56 if the tracks were independent, and about 6 tracks (three calls, roughly 54 credits at the roughly 18 credits per call observed here) reach 90%. These numbers are an illustration, not a result: tracks from one call are probably correlated, so they are optimistic, and the application deliberately takes the first finished track and does not run this loop.

RQ5 — How do the AI components change the specification?

> *Scenario.* A user with a Gemini key asks the app to "predict the best music style" for Hongdae from its photo and description. Does the specification change, and is the change an improvement?

What was done, and what the evidence is. Gemini (model *gemini-3.5-flash-lite*) is used in three implemented features: persona chat, a photo-and-text style predictor, and a per-language accessibility estimate. The request format was validated against Google's endpoint (a deliberately invalid key returned an authentication error, not a format error), and the model identifier was re-verified against Google's model pages on 20 Sep 2026. The current style predictor receives the application's tempo, tonal center, mode and vocal treatment as fixed inputs and returns only genre, instrumentation, mood descriptors, a one-sentence arrangement and a rationale (temperature 0.35); the prediction is cached per place and mapping version. The repository holds no recorded output from a run with an API key (this does not mean a run is impossible, only that none has been saved), so the comparison below is a *hand simulation*: the function's own instructions applied by hand to the real photo and text, for the earlier version of the predictor, which also chose the key [project reports, 13 Sep 2026]. Because the current predictor cannot change the key, the disagreement in Table 9 can no longer arise through this path; what it shows is the input-level limitation of a regional value.

Table 9. Rule-based mode versus a photo-grounded hand-simulated prediction from the earlier predictor, five landmarks chosen to span score bands and categories. Hand-simulated predictions are not recorded Gemini output.

Landmark	Rule-based key (deciding rule)	Hand-simulated photo-based key	Agree
Hongdae	minor (regional Pleasantness default)	major (photo shows a drizzly daytime street)	✗
Gyeongbokgung Palace	major (traditional override)	major	✓
5.18 National Cemetery	minor (solemn override)	minor	✓
Haeundae Beach	major (nature override)	major	✓
POSCO Pohang Steelworks	major (regional default, exactly at the median)	major	✓

Four of five agreed. The one disagreement is traceable: the rule reads a regional survey value, the photo shows one afternoon.

The language-accessibility estimator is display-only: a code search found no reference to it in the score, key or prompt paths, so choosing a language changes what a visitor reads and not what they hear. A hand-simulated English estimate for four provinces (Seoul ~78, Incheon ~68, Busan ~62, Jeju ~52) preserved the order of the 2024 EF English Proficiency Index values cited in the project (550, 532, 520, and below the national average for Jeju); estimates for other languages have no external anchor.

Implementation audit. Reading the integration code showed three defects that would have affected any live Suno or Gemini use. Each was fixed and checked in a browser with a mocked network layer (no credits spent, no real keys). (i) The Suno key, and on pages opened from disk the Gemini key placed in the web address, passed through a public third-party relay; both providers answer browser preflight requests (checked for a local web server and for a page opened from disk), so calls are now direct and the Gemini key travels in a request header. (ii) Live generation rebuilt the rule-based prompt and ignored the AI-assisted prompt shown in the panel. (iii) A partial streaming address could be returned before the track finished.

RQ6 — How does generated audio lead to listener-perceived Eventfulness?

> *Scenario.* A listener hears an unlabelled 25-second clip and rates eight adjectives. Do clips generated for higher-activity places sound more eventful?

No data have been collected. The protocol, a blinded static rating form, a clip-preparation script and an analysis script exist (Methods), and 20 unit tests check the formulas on synthetic fixtures only. The stimulus manifest is not yet marked ready and every clip's key review is marked pending, so no participant has been invited. The primary result will be the Spearman correlation between the ten generating scores and the ten clip-mean adapted-Eventfulness ratings. Because round 1 and round 2 differ in prompt and request shape, the clip set to be frozen must be chosen before recruitment, and the generation results above set the limit on what a null result could mean: if the audio does not carry the specified tempo or mode, a flat listener result would not distinguish a failure of perception from a failure of generation.

RQ7 — How does the model relate to the actual places?

> *Scenario.* A person stands in Hongdae at night and in Gyeongbokgung at noon and rates the same eight adjectives. Does the Activity Proxy predict what they report?

No data have been collected. The field protocol, a 47-column observation template (header row only), a Korean questionnaire, a printable field pack and an analysis script (t-interval summaries per site-time) are ready. It is the only planned test of whether the model, rather than its musical rendering, describes real places. The two published reference regions (RQ1) are the only external evidence that currently touches this link.

RQ8 — How does the interface lead to accessible exploration?

> *Scenario.* A blind visitor with a screen reader tabs across the map. Do they learn which regions are calm, and can they do it without help?

Implemented, not user-tested. All 17 regions are keyboard tab stops whose accessible labels give the name, score, mood label, the underlying inputs and evidence status; a single polite live announcement reports changes; focusing a region plays its score as a pitch on a pentatonic scale, stretched across the observed range because the proxy compresses scores into roughly 33–61, with the region's east–west map position mapped to left–right stereo placement; the four arrow keys move to the nearest region in that compass direction; a text-and-geometry panel gives deaf and hard-of-hearing visitors the same parameters as the prompt; untranslated text is marked with its own language; and motion respects the user's reduced-motion setting. These are implemented and were checked by the development team as technical behavior. They have not been tried by blind or low-vision users, so effectiveness and usability are unknown, and sighted people with eyes closed would not substitute for them.

Discussion

The project's central claim is that regional data can be turned into music that conveys a place's character. This report found that the chain supporting that claim is intact in structure but uneven in strength (Table 10). Its first four links can be computed or measured today; what they show is more cautious than the interface language. The single-listener Sejong diagnostic demonstrates that automatic and perceived tempo can diverge, but it does not replace a formal listener study.

Table 10. Summary of the chain by evidence class.

RQ	Link	Evidence	n	Main finding	Main caveat
RQ1	Regional data → score	computed	17 regions	Order is not robust to weights (ρ median 0.87); one external sanity check (n = 2)	Weights uncalibrated
RQ2	Region → landmark	computed	75 landmarks	Region explains 12.2%; offsets are authored	Illustrative scores
RQ3	Score → specification	computed	75 landmarks	Pleasantness decides 46.7% of modes, in 17 regional decisions	Design assumption untested
RQ4	Specification → audio	measured + single-listener diagnostic	20 tracks plus follow-ups; 1 listener, 1 clip	Mode 5/10 then 7/10; Sejong automatic 95.3 vs human 48 BPM; 96 BPM after correction	One place, one listener, sequential edits
RQ5	AI components	hand-simulated	5 + 4 cases	4/5 key agreement; no live model call	Not Gemini output
RQ6	Audio → listener	pending	0 participants	—	Protocol ready
RQ7	Model → real places	pending	0 observations	—	Template empty
RQ8	Interface → access	pending	0 users	—	Technical checks only

Data to score: a plausible ordering with fragile details

Seoul, Busan and Jeju stand out and Gwangju is lowest, and the one external reference points the right way. But the ordering below the top is sensitive to the weights, and the weights are not what they appear to be: the component with a 30% weight had no rank association with the score. Using more than one aggregation, reporting rank ranges instead of ranks, or calibrating weights against field data (RQ7) would be more defensible than a single set of hypothesised weights. The Visitor Pleasantness composite compresses 17 regions into a 6.7-point band, so using it as a major/minor switch turns small survey differences into a categorical difference in how places sound.

Score to place: authored offsets carry the landmark

Because region explains about an eighth of landmark-score variance, the sound a visitor hears for a landmark is mostly a judgment made by the author of the offset. This is disclosed in the interface, but it limits what any listener or field result for a landmark could say about the official statistics. It also explains why the popularity index, which is independent of the offsets, is only weakly related to them.

Specification to audio: requests are soft hints, and some are hard to hear

The generator honored the requested mode in about half to two-thirds of clips; in the two main rounds it never returned minor when asked (five requests at three landmarks), although follow-up generations did so under some prompt conditions. The requested tempo was usually detectable in the audio even when a single estimate disagreed. Two differences from the intended design are worth stating plainly. First, mode is controlled less by the mode words than by the genre words around them, and not predictably (Gangnam responded, Asan did not). Second, a pitch-level request (root note) was not reliably honored and is now phrased as a soft hint. The investigation also exposed two defects in the project's own code and data, a keyword collision and a missing real-world fact, which changed outcomes more cleanly than any prompt rewording. This points to checking the project's own category logic before adjusting a third-party model.

The Sejong case adds a third distinction: *automatically detected BPM* is not necessarily the *BPM a listener selects as the main pulse*. The round-2 source appeared successful at 95.3 BPM automatically but was tapped at 48 BPM; even an overlaid 95 BPM pulse left the 47 BPM larger beat intact. Only correction of the complete time axis and the conflicting hi-hat produced a 96 BPM tap. Where exact tempo carries the data meaning, a reproducible architecture is therefore to use generated audio as harmonic and textural material, while enforcing tempo with deterministic retiming and rhythm layers rather than treating the generator as the final timing authority.

AI components: implemented and constrained, outputs not yet recorded

The earlier photo-grounded style predictor disagreed with the rule path where a regional value is likely to be wrong about a particular scene. The current predictor takes the mode as a fixed input, so it cannot repeat that disagreement; whether photo or place-level evidence should be allowed to set the mode is now a design choice,

not something the model does by default. A key-backed run recording the predictor's genre, instrumentation and arrangement on the same cases would show how much it adds beyond the rule-based style.

What a listener or field result would mean

If listeners rank the clips in the generating order (positive ρ, pairwise accuracy well above 0.5), that supports the claim that the generated audio carries an activity ordering that people can hear on these clips; it would say nothing about the places. If it does not, the RQ4 results mean that the cause cannot be assigned to perception rather than generation without a human check of tempo and mode in the same clips. A positive field result in RQ7 would support the Activity Proxy; a null would say the official-statistics composite does not track perceived Eventfulness at the sampled sites.

Limitations and future pathways

- **Small, single-provider audio sample.** 20 tracks in two rounds plus about 20 follow-ups, one provider and one model version, one generation per prompt in most cells. Only the tempo of one Sejong clip was checked informally by one person; all remaining modes and tempos are automatic detections. Intervals for 5/10 and 7/10 are wide (RQ4).
- **Post hoc follow-ups.** The genre-word and request-shape comparisons were not preregistered, and changed more than one thing at a time (for example the custom request also dropped the scenic sentence). They suggest hypotheses.
- **Tempo diagnostics use the request.** They show the requested pulse is detectable, not that it is perceived. Sejong's 95.3-versus-48 BPM discrepancy illustrates this limitation, but n = 1 cannot estimate its prevalence.
- **Hypothesised weights and an authored landmark layer.** No weight is fitted; no landmark is measured.
- **Provenance of two regions.** The inputs for Seoul and Daejeon are in the model but not in the 15-region source file; their sources need reconciling before submission.
- **Gemini outputs not recorded.** The three Gemini features are implemented, their request format and model identifier were checked, and the style predictor is constrained to the application's fixed parameters, but the repository holds no recorded output from a key-backed run. The comparison in RQ5 is a hand simulation of the earlier predictor. Any key-backed outputs should be added.
- **No formal participant study.** The single informal Sejong tap diagnostic had no repeated trials, multiple listeners or demographic data; it is an engineering check, not RQ6 data. Listener perception, in-situ perception and intended-user accessibility remain untested. Korean-language and cultural effects on mode–valence associations are unexamined.

Next steps, in order of value per effort: (1) have multiple independent listeners tap every clip blind, have a musician identify mode, and replicate the corrected Sejong clip's 96 BPM result; (2) repeat each prompt several times before claiming any prompt change works; (3) test the genre-word hypothesis at more landmarks; (4) freeze one clip set and run the listener experiment with at least 15 and a target of 24 adults; (5) run the field pilot at 6–10 sites with 2 time windows; (6) run the Gemini checks with a live key; (7) when possible, study the interface with blind and low-vision users; (8) use field data to replace hypothesised weights with calibrated ones, and replace authored landmark offsets with observations where available.

Conclusion

Moodscape's pipeline can be traced end to end, and its first four links are implemented and measurable. As of now: regional ordering is plausible but weight-sensitive; landmark scores are mostly authored; the specification's mode rule is regional and coarse; and the generator does not reliably guarantee either requested minor mode or the pulse a person perceives. At Sejong, an automatic 95 BPM match was heard at 48 BPM and reached 96 BPM only after deterministic retiming and hi-hat replacement. Whether listeners hear the intended ordering across

the clip set, whether the model describes real places and whether the interface helps people who cannot see the map remain open questions with prepared tests and no formal results.

Methods

Study design and evidence classes

This is a system-validation study, organized as one analysis per link in the chain (Fig. 1). Analyses RQ1–RQ5 use data already in the project repository and measurements of generated audio. RQ4 additionally includes an informal blind tap of one Sejong clip by the same single person as a sequential engineering diagnostic; it had no sampling protocol or inferential analysis and is not counted as RQ6 participant data. The formal RQ6–RQ8 studies are specified here but have no data.

Regional model (RQ1)

Nine official inputs per region (17 regions) are used: 2025 population and population density (Statistics Korea), 2024 domestic tourism trips and visitor survey measures (overall satisfaction, recommendation intention, revisit intention, crowding satisfaction; Korea National Tourism Survey 2024), and 2023 noise/vibration complaints and regulated noise-emitting facilities (Ministry of Environment). Five activity components are derived: density; tourism trips per resident; complaints per 100,000 residents; facilities per 100,000 residents; crowding pressure (100 − crowding satisfaction). Each is transformed with *log1p*, min–max scaled across the 17 regions, and combined as *100 × (0.35 density + 0.30 tourism + 0.15 complaints + 0.10 facilities + 0.10 crowding pressure)*. Visitor Pleasantness is *0.40 overall satisfaction + 0.25 recommendation + 0.20 revisit + 0.15 crowding satisfaction*. Weights are stated hypotheses, not fitted coefficients. The Activity Proxy is not an ISO 12913 coordinate. The 15-region source file omits Seoul and Daejeon, whose published soundscape studies are tracked separately.

Statistics. Descriptive statistics use the sample SD. Component influence is the Spearman correlation of the scaled component with the final score. Weight sensitivity used 10,000 Dirichlet draws (concentration 20 around the stated weights, and a flat Dirichlet) with seed 20260920; rank intervals are the 2.5th–97.5th percentiles of each region's rank.

Landmark layer and popularity (RQ2)

Each of 75 landmarks has an authored score; at load, the score is shifted so the region's authored baseline equals its computed Activity Proxy, and clipped to [0, 1]. Variance partition used a one-way random-effects model by region ($\eta^2 = SS_{between}/SS_{total}$; ICC(1) from the ANOVA mean squares with the unbalanced-design group size). Readership: English Wikipedia monthly *user* pageviews, all access, 1 Sep 2025 – 31 Aug 2026 [13]. The index is *100 × (log1p v − min)/(max − min)* over complete records; missing and partial records are not scored. Association with the landmark score is Spearman ρ with a permutation p-value.

Musical specification (RQ3)

For a landmark score $s \in [0, 1]$: tempo $T = round(42 + 110\ s)$ BPM; pitch $f = 440 \cdot 2^{k/12}$ Hz with $k = round(12\ s)$ semitones above A4, and the root letter is the pitch class k semitones above A (A, A♯, …, G♯). Mode is decided in order: *minor* if the place is solemn (memorial, cemetery, observatory, cave); else *major* if it is traditional, natural or administrative (hanok, temple, palace, park, beach, government and similar); else *major* if $P_r \geq median(P)$ and *minor* otherwise, where P_r is the region's Visitor Pleasantness and the median is over the 17 regions. Style words come from category and score band; for a minor key a fixed list of positive-valence words is removed, and the mode is stated in an early sentence (Appendix A). A vocal directive keeps most places instrumental. The text sent to the generator is this style string plus a title, in the provider's separate style-and-title request shape. The rules were evaluated by running the project's own implementation, so the tables reflect it and not a reimplementation.

Audio generation (RQ4)

Audio was generated with Suno through the `sunoapi.org` service, model V4_5, between 13 and 20 September 2026. Round 1 sent the original prompt in the non-custom request shape for ten landmarks chosen at rank quantiles of the 75 scores (score 0–87) with Gangnam substituted to include a minor-key case. Round 2 generated the same ten again, as new audio, with the revised prompt builder and the custom request shape. Follow-up experiments (reworded prompts, an explicit root, genre-word swaps, three Sejong variants, and a small pilot of three API calls) used the same service. One API call usually returns two tracks; the first returned track was analyzed in each round, and both tracks were analyzed in the pilot. Each generation was kept regardless of outcome. Provider task identifiers are recorded for the pilot; they are not in the project's metadata files for earlier generations. The retrieval scripts accept the first audio URL the service lists, which could in principle be a streaming URL taken before a track is complete (the pattern fixed in the app, RQ5); analyzed durations were 71–182 s, and the analysis is of the audio as downloaded.

Audio analysis (RQ4)

All clips were analyzed again with one procedure (librosa 1.0.0 [9]), using the same audio files and no new generation; audio was read as mono at its native sampling rate. **Key.** Let c be the time-averaged, standardized constant-Q chroma vector and q_m the Krumhansl–Kessler tone profile for mode $m \in$ {major, minor} [8]. For each root $k = 0, \ldots, 11$ compute $r(k, m) = corr(c, q_m \text{ rotated by } k)$. The reported key is the pair (k, m) with the largest r, and that largest r is reported as the key-fit score. It is a correlation, not the probability that the key is correct. **Tempo.** The single-value estimate is the tempo at which the autocorrelation of the onset-strength envelope, weighted by a log-normal tempo prior centered at 120 BPM, is largest; the earlier reports used the beat tracker built on the same envelope, and the two can differ (for example 44.3 versus 89.3 BPM for one round-1 clip). For the diagnostic, $A(\tau)$ is the time-averaged autocorrelation tempogram; its local maxima between 30 and 300 BPM are ranked by height, and a request q BPM is counted as present if one of the three highest peaks b satisfies $|b - q| / q \le 0.05$. The null distribution shuffles the ten requests among the ten clips 20,000 times and the p-value is $(1 + \#\{\text{null} \ge \text{observed}\}) / (1 + 20{,}000)$. Rank agreement is Spearman's $\rho = 1 - 6\Sigma d^2 / (n(n^2 - 1))$ without ties, with average ranks and Pearson correlation of ranks when there are ties. Success rates carry Clopper–Pearson intervals [10]; contrasts in Fig. 10 use Fisher's exact test. Key detection is imperfect on sparse or ambient music and cannot separate relative major and minor keys perfectly. Apart from the single Sejong case below and earlier informal listening that identified its style mismatch, no human verified the clips.

Sejong perceptual-tempo diagnosis and correction. On 25 October 2026 one informal listener tapped the primary pulse of anonymous 25-second files. This was a sequential engineering diagnostic of metrical-level error, not a recruited participant study. The round-2 Sejong source and a strict-prompt regeneration were each tapped at 48 BPM; the source with kick and rim placed every 0.6316 seconds was tapped at 47 BPM. For the final version, the human source measurement of 47 BPM was converted to the 95 BPM target by a 95/47 = 2.021× pitch-preserving librosa phase-vocoder time compression. A sixth-order Butterworth low-pass split at 5.2 kHz retained 12% of the original high band to suppress the conflicting hi-hat; a seeded synthesized closed hi-hat was placed once per quarter note, kick on every quarter note and rim on beats 2 and 4. Pitch was preserved and duration changed from 164.413 to 81.341 seconds. The same listener tapped the new anonymous 25-second file at 96 BPM. The decision rule was absolute error within ±5% of target; observed error was 1.1%. Source, intermediate and final audio, hashes, parameters and scripts are retained in the repository.

AI components and implementation audit (RQ5)

The Gemini model was *gemini-3.5-flash-lite*, re-verified against Google's model pages on 20 Sep 2026. Three features use it: persona chat; multimodal style prediction from a place's photo and text, in which the application's tempo, tonal center, mode and vocal treatment are supplied as fixed inputs and the model returns genre, instrumentation, mood descriptors, an arrangement sentence and a rationale as structured output (temperature 0.35), cached per place and mapping version; and a per-language accessibility estimate. The request format was validated against Google's endpoint with an invalid key (an authentication error, not a format error). No recorded key-backed output exists, so the comparison of the earlier predictor with the rule path used a hand simulation: each

function's instructions applied by hand to real photos and text. The integration audit read the Suno and Gemini code paths, probed browser preflight behavior on both services and replayed the fixed paths against a mocked network layer.

Interface and accessibility implementation (RQ8)

Each region is a keyboard target with an accessible label; one polite live announcement reports selections; a synthesized tone encodes the focused region's score and east–west position; arrow keys move to the nearest region in that compass direction; a text panel states the generated music's parameters; untranslated strings declare their language; and animation respects the reduced-motion setting. Checks by the development team are technical only.

Planned listener experiment (RQ6)

Ten 25-second clips (5–30 s of each source, loudness-normalized to −18 LUFS, −1 dBTP, metadata stripped, SHA-256 hashes frozen) are presented in a shuffled order, without names or scores, to adults recruited through the researcher's contacts. Participants rate eight descriptors (pleasant, chaotic, vibrant, uneventful, calm, annoying, eventful, monotonous) on a five-point agreement scale. Adapted Eventfulness is $[(\text{eventful} - \text{uneventful}) + \cos 45^\circ(\text{chaotic} - \text{calm} + \text{vibrant} - \text{monotonous})] / (4 + \sqrt{32})$, following ISO/TS 12913-3 [4]; Pleasantness uses the analogous formula. Target 24 complete participants (minimum 15 for an exploratory pilot). Primary analysis: Spearman correlation between the ten generating scores and ten clip means, two-sided permutation test (49,999 permutations), participant-bootstrap interval (5,000). Prespecified secondary: pairwise ordering accuracy over 45 clip pairs (chance 0.5). Descriptive: Lin's concordance correlation [12] and WAPE/MAE on min–max-scaled series, ICC(2,1)/(2,k) [11] and Kendall's W. These formulas were checked against published examples and hand calculations in 20 unit tests that use synthetic fixtures only. The protocol was amended on 20 Sep 2026, before any data, to add the secondary and descriptive metrics.

Table 11. Prespecified listener-experiment analyses (RQ6). None has been run on participant data.

Role	Quantity	What it answers	Decision rule
Primary	Spearman ρ, clip means vs generating score; permutation p	Is the ordering preserved?	$\rho > 0$ and $p < .05$, else report non-support
Secondary	Pairwise ordering accuracy (45 pairs) with bootstrap interval	Plain-language match rate	None; chance = 0.50
Descriptive	Lin's CCC, WAPE, MAE on min–max-scaled series	How closely the shapes coincide	None; scale-dependent
Descriptive	ICC(2,1), ICC(2,k), Kendall's W	Do listeners agree with each other?	None

Planned field study (RQ7)

Following the ISO/TS 12913-2 questionnaire approach [3] as applied in [1], the pilot would sample 6–10 sites (commercial street, residential street, park or natural area, transit area, market or heritage area) in one or two provinces, in two time windows each, with about five independent raters per site-time. Raters complete the same eight-item questionnaire; the recorder logs sound level, pedestrian and vehicle counts, location, weather and land use. Ratings are converted to Pleasantness and Eventfulness, aggregated by site-time with t-intervals, and compared with the Activity Proxy, with grouped analysis (the unit is the site-time, not the rating). The observation template has 47 columns and no data rows; the Korean questionnaire needs a back-translation pass before use. Adult participants only; the supervising institution's review is required before recruitment.

Software, reproducibility and ethics

Analyses were run in Python 3.13 with NumPy, SciPy, Matplotlib and librosa 1.0.0. Random seeds are fixed and recorded. All numbers in this report are recomputed from files in the repository by the analysis scripts listed under Data and code availability, which also draw every figure. No personal data were collected. The listener and field studies will need institutional review before they begin.

Use of AI tools

The system uses Gemini (style prediction, chat, language estimate) and Suno (audio generation) as components. The analysis scripts, parts of the system code and this manuscript were prepared with the assistance of Claude (Anthropic) working in the project repository under the direction of the project lead, who made the research decisions; every number in the text was recomputed from repository data, and the authors remain responsible for the content.

Appendix A — Example specification

The text the current code builds for Gangnam (requested minor, 115 BPM), shown as sent in the style field of a custom-mode request. The title is “Gangnam — Moodscape” and the request is instrumental.

```
Music for a glittering high-rise district of designer boutiques and rooftop bars. In a minor key
(around F) — melancholic, somber, shadowed tonality, no major-key brightness. lo-fi hip-hop,
relaxing city-pop synth pads, urban beat, around 115 BPM. Instrumentation: glossy K-pop synth-pop,
sidechain-pumped bass, glass-tower shimmer. Cinematic, atmospheric, no vocals, no lyrics.
```

Data and code availability

All inputs and scripts are in the project repository (`github.com/clairenah19/sound-moodscape`). Data: `research/moodscape_15_regions_official_proxies.csv`, `moodscape_2024_national_tourism_province_perception.csv`, `moodscape_landmark_pageviews.csv`, `listener_stimulus_metadata.csv`, `audio_analysis_results.csv`, `audio_analysis_results_v2.csv`, `prompt_ablation_results.csv`, `paper_audio_reanalysis.json`, `paper_model_export.json`, `listener_clips/clip07_sejong_retimed_hihat_fixed.json`. Code: `data.js` and `prompt.js` (model and prompts), `research/paper_analysis.py` (every statistic and figure), `research/paper_audio_reanalysis.py` (audio measurements), `research/retime_to_perceived_bpm.py`, `research/fix_hihat_tempo.py` and `research/add_deterministic_pulse.py` (Sejong tempo correction), `research/analyze_listener_experiment.py` and `research/analyze_soundwalk.py` (planned analyses). Generated and corrected clips are in `research/listener_clips/` and remain subject to the provider's terms. Third-party photographs keep their own licenses. No formal participant-level data exist yet.

Author contributions, funding and competing interests

[To be completed by the authors.] Author names, affiliations, a corresponding author, funding and a competing-interests statement are pending.

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Entries 1–5 and 13 were checked against the reference paper's own reference list or the project's documentation. Entries marked † were entered from the project's source catalog or the author's knowledge and have not been re-checked against the publisher; verify each before submission.